King Saud University

College of Business Administration

Management Information System Department

MIS431--Business Process Management

Supervised by: Dr. Athary Al wasel

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Final Full Report

ID	Student Name
444201362	Ghadir Almosa
444201234	Aljohara Alkaridis
444201754	Reem Alotaibi
444201703	Hissah altukhies
444201278	Dania AL,Qahtani
444201550	Lama Alqahtani

Team Members' Work Contributions:

Student Name	Work
Dania AL,Qahtani	• Write Highlight the Important Processes for Improvement • Handle and Draw the Exception Process • Root-cause analysis+ why why Diagram • Quantitative processes analysis • Key findings • Conclusion • References
Hissah altukhies	• Process identification • Process discovery • Worked though the whole To-Be -Diagram modifications and improvements • Key Issues & Redesign Goals. • Link Between Issues, Redesign Goals, and Applied Heuristics
Aljohara Alkaridis	Describe the business problem. • Sub-process • Executive Summary • SMSA Shipping Case Study • Root-cause analysis • Why Why Diagram • Quantitative processes analysis Conclusion
Reem Alotaibi	• Worked though the whole To-Be -Diagram modifications and improvements • Link Between Issues, Redesign Goals, and Applied Heuristics • Key Issues & Redesign Goals • Drawing as-is model and Discovery Methods • List of processes using the process architecture
Lama Alqahtani	• Identify the critical (chosen) process. • Thinking through the entire scenario • Drawing the entire scenario on paper (without exceptions or sub-processes) • Meeting with Ms. Athari • Updating the plan • Organizing all activities and elements to ensure a clear and accurate plan • Reviewing some rules • Qualitative analysis Identifying problems and Proposing solutions • Designing a proposed redesign for discussion and further adjustments • Modeling the sub-process
Ghadir Almosa	• Writing the organization overview • Identifying the process selected, the process scope and the main participants. • Drawing the As-Is diagram • Worked though the whole To-Be -Diagram modifications and improvements • Key Issues & Redesign Goals. • Heuristics • Link Between Issues, Redesign Goals, and Applied Heuristics • Alternative redesign options considered but not adopted
(All updates and presentations were done by all team members.)	

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Organization Overview:

SMSA Express SMSA Express is a leading Saudi Arabian courier service and logistics firm. Since 1994, the company has been providing end-to-end logistics solutions ranging from domestic and international delivery, freight forwarding, warehousing, e-commerce logistics, and last-mile delivery. With its vast network of operations in all regions of the Kingdom, SMSA has gained a reputation for being dependable, and quality-based in delivery. It is both corporate and individual client-server and offers customized delivery services with state-of-the-art tracking facilities and extensive transport reach. The core goal of SMSA is to offer efficient, secure, and technology-enabled logistics solutions that meet the changing requirements of the Saudi market. Since the logistics industry is a competitive industry, SMSA always attempts to enhance its business processes so it can provide more quality services and enhanced operational efficiency. To the interest of the project, SMSA Express has been chosen as the target company whose business process needs to be studied and enhanced to improve overall efficiency, coordination, and service reliability for its clients.

Business Problem description:

SMSA Express is a logistics and courier service company that specializes in domestic and international shipping and delivery solutions. Despite its established presence in the Saudi market, the company is currently facing significant challenges in its delivery process, which directly affects customer satisfaction and operational efficiency. One of the main problems is the frequent delays in delivering shipments, especially during peak seasons or holidays. These delays are primarily caused by several internal process inefficiencies. Firstly, shipment data is often entered manually, which increases the risk of human error and leads to incorrect or incomplete information being recorded. Secondly, there are delays in updating shipment status in the system, preventing customers and staff from tracking deliveries in real time. Additionally, weak communication and coordination between the operations department and field drivers further complicate the delivery schedule, resulting in missed delivery windows. Lastly, the accumulation of a large number of orders during high-demand periods overwhelms the current process and causes further delays. These challenges highlight the need for process improvement to enhance coordination, automate manual tasks, and improve communication among stakeholders. Addressing these issues will not only streamline the delivery process but also improve customer satisfaction and strengthen SMSA Express’s competitive position in the logistics industry.

List of Processes Using the Process Architecture

SMSA Express operates through several key processes that ensure shipments are handled efficiently from order placement to final delivery. The following list presents the main processes involved in the company’s logistics and delivery operations:

1. **Order Placement:** Customers create shipment orders through SMSA’s online platform or service centers.

2. **Shipment Data Entry:** Shipment details are entered into the system, including sender and receiver information, package weight, and delivery location.

3. **Package Pickup:** The assigned driver collects the package from the customer or company location.

4. S**orting and Dispatching:** Packages are sorted at regional hubs based on destination and type of delivery.

5. **Transportation:** Shipments are transported between cities or regions using SMSA's network of vehicles.

6. **Out for Delivery:** The delivery driver receives the packages and prepares for final delivery to customers.

7. **Delivery to Customer:** Packages are delivered to the recipients, and proof of delivery is recorded.

8. **Status Update and Confirmation:** The system is updated with delivery status, allowing customers and staff to track shipments in real time.

Important Processes Highlighted for Improvement:

Based on the process architecture of **SMSA Express**, several processes require improvement to enhance efficiency and customer satisfaction.

1.Shipment Data Entry:

This process is still performed manually in many cases, leading to delays and human errors when entering shipment details such as addresses.

2.Out for Delivery:

During this process, poor route planning and lack of real-time communication with drivers cause delivery delays and reduce efficiency.

3.Delivery to Customer:

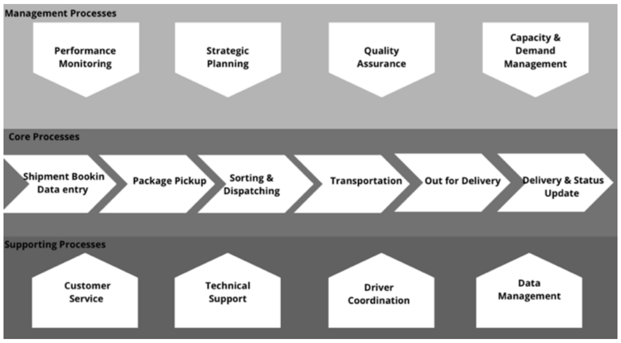
Inconsistent delivery times and lack of delivery confirmation negatively impact customer satisfaction and trust in the service.

4.Status Update and Confirmation:

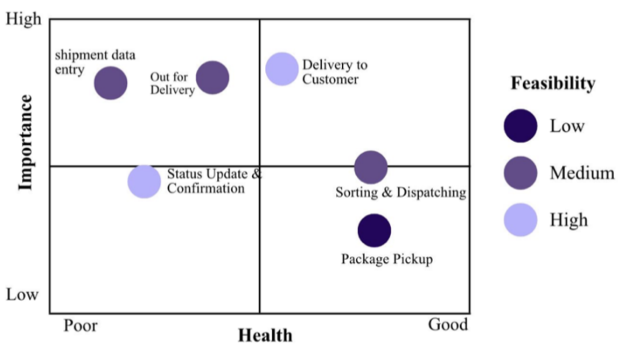
Delays in system updates affect shipment tracking. Enhancing real-time tracking visibility improves transparency and customer experience.

Process Identification

Process Architecture:



Process Portfolio:



After analyzing the core processes of SMSA Express using the process portfolio framework, the “Out for Delivery” and “Shipment Data Entry” processes were identified as the most critical processes requiring immediate improvement.

Identify the criteria (chosen process)

The process we are discussing is the “Delivery Process” at SMSA Express.

It was selected based on the following criteria:

1- Impact on Customers:

It directly influences the customer satisfaction and the company image and reputation. delays or errors affects the overall customer experience in a negative way.

2-Process Frequency:

Frequent and critical operations within the company, during peak seasons and holidays especially.

3-Existing Issues:

The process has many problems and challenges for example shipment delays, inaccurate data entry, and weak coordination between departments.

4-Cross-Functional Nature:

It contains many departments including customer service, warehouse, drivers, and logistics, and all of them need to be aligned for improving overall efficiency and workflow.

5-Improvement Potential:

There’s great potential for automation and digital enhancement to reduce errors made by humans and improve communication and tracking systems.

Assumptions

1.The data entry for shipments is mainly handled manually by one employee in each branch.

2.Drivers communicate with the operations department primarily through phone calls or messages.

3.There is a tracking system in place, but it is not updated regularly or taken seriously by the staff.

4.Most delivery delays occur during weekends and peak seasons due to high demand.

5.The company has sufficient resources and technical capability to implement digital process improvements.

6.Customers prefer real-time tracking of their shipments through the company’s website or mobile application.

7.Departments such as customer service, warehouse, and delivery work independently without full system integration.

8.Employees are not fully trained on using the system efficiently, which causes data entry errors and tracking delays

9.The company lacks an effective system or strategy to manage a high volume of orders during peak seasons.

Process Selected:

The choice of the process for this project is Delivery Process at SMSA Express. This is one of the most fundamental activities conducted by the company and impacts customer satisfaction directly and is thus an area worth focusing on for improvement. This particular task is completed each day and is viewed as one of high priority workflow because of its frequency and business impact.

The Delivery Process:

This is the sequence of activities needed to deliver or move the shipment from SMSA to the destination customer. This is one of the most crucial processes because it is the last point of customer interaction and impacts their perceptions of service quality to a large extent. It was found from Phase 1 results that several flaws exist at this point as well: lack of punctuality, laborious status updates, poor communication processes for drivers, and peak season pressures.

Main Participants:

The key participants who take part in this whole procedure include

1. **warehouse staff**, who deal with receiving and unloading shipments.
2. **operations personnel**, responsible for allocating drivers for delivering goods to customers.
3. **drivers**, whose task is to carry out delivering processes and enter its status accordingly.
4. **customer and receiving personnel,** responsible for delivering goods to them.

Process Scope:

The Delivery Process begins as soon as the sorted packages are ready to be delivered from the warehouse to the respective drivers for delivery to their destinations and stops as soon as the packages are delivered successfully and the successful delivery is entered into the system or as soon as they are returned to the warehouse for cases where the customer is not available to take reception of packages at his/her destination.

Discovery method

1-Interviews

For the first discovery method, we selected Interviews. This method is suitable for understanding real workflow details, challenges, and communication issues directly from the people who work on the delivery process at SMSA Express. Interviews provide deeper insights into how the process actually operates compared to how it is documented.

How the Method Was Used:

We conducted a structured interview with an Operations Supervisor and a Field Driver, since both roles are directly involved in the delivery process. The interview included open-ended questions to understand the workflow, daily operations, and common issues.

The questions focused on:

• How the delivery process starts inside the warehouse after sorting

• How drivers receive assignments and plan their routes

• How shipment information is recorded and updated

• How communication occurs between operations staff and drivers

• Common causes of delivery delays

• Challenges during peak seasons

• How shipment status is tracked and updated in the system

Information Collected:

From the interviews, we gathered the following information:

• Shipment data is still entered manually at several stages, which causes errors and slows down operations.

• Updating shipment status is sometimes delayed, affecting real-time tracking for customers.

• Communication between operations and drivers is not fully efficient, especially during peak times.

• Drivers sometimes receive incomplete or unclear information about routes or customer details.

• High volumes of shipments during holidays overwhelm the current process.

• There is no unified automated system that connects all teams seamlessly.

This information helped clarify the actual steps and problems inside the process beyond what is assumed or documented.

Key Insights:

• Manual data entry is one of the main sources of delays and errors.

• Communication gaps between departments cause missed delivery windows.

• Real-time tracking is affected by delays in updating shipment status.

• The process becomes highly inefficient during peak seasons due to lack of automation.

• Employees agree that improving system integration and automating tasks would significantly enhance delivery speed.

2- Observation

For the second discovery method, we selected Observation. This method is suitable for directly seeing how the delivery process works at SMSA Express in real time. Observation helps identify gaps between the documented workflow and the actual daily operations by watching employees perform their tasks in the warehouse and during the delivery process.

How the Method Was Used: We conducted a structured observation inside the warehouse and at the dispatch area where drivers receive their shipments before starting their routes.

The observation focused on the following points:

• How shipments are sorted and prepared for delivery

• How drivers receive their daily assignments

• How shipment details are reviewed, confirmed, and loaded

• How operations staff communicate instructions to drivers

• How delays or issues are handled in real-time

• How shipment information is updated in the system during dispatch

Information Collected:

From the observation, we gathered the following information:

• Shipment sorting takes longer when volumes are high, which pushes delivery start times later.

• Drivers often wait for final confirmation of shipments, causing delays before they leave the warehouse.

• Communication between operations staff and drivers happens both verbally and through mobile devices, but sometimes instructions are unclear.

• Some steps are skipped during peak hours to save time, which affects data accuracy.

• Manual checking of shipment details takes extra time and introduces errors.

• Updating shipment status is sometimes delayed because drivers focus on loading quickly to start their routes.

Key Insights:

• Real-time operations differ from the documented procedures due to time pressure and manual dependencies.

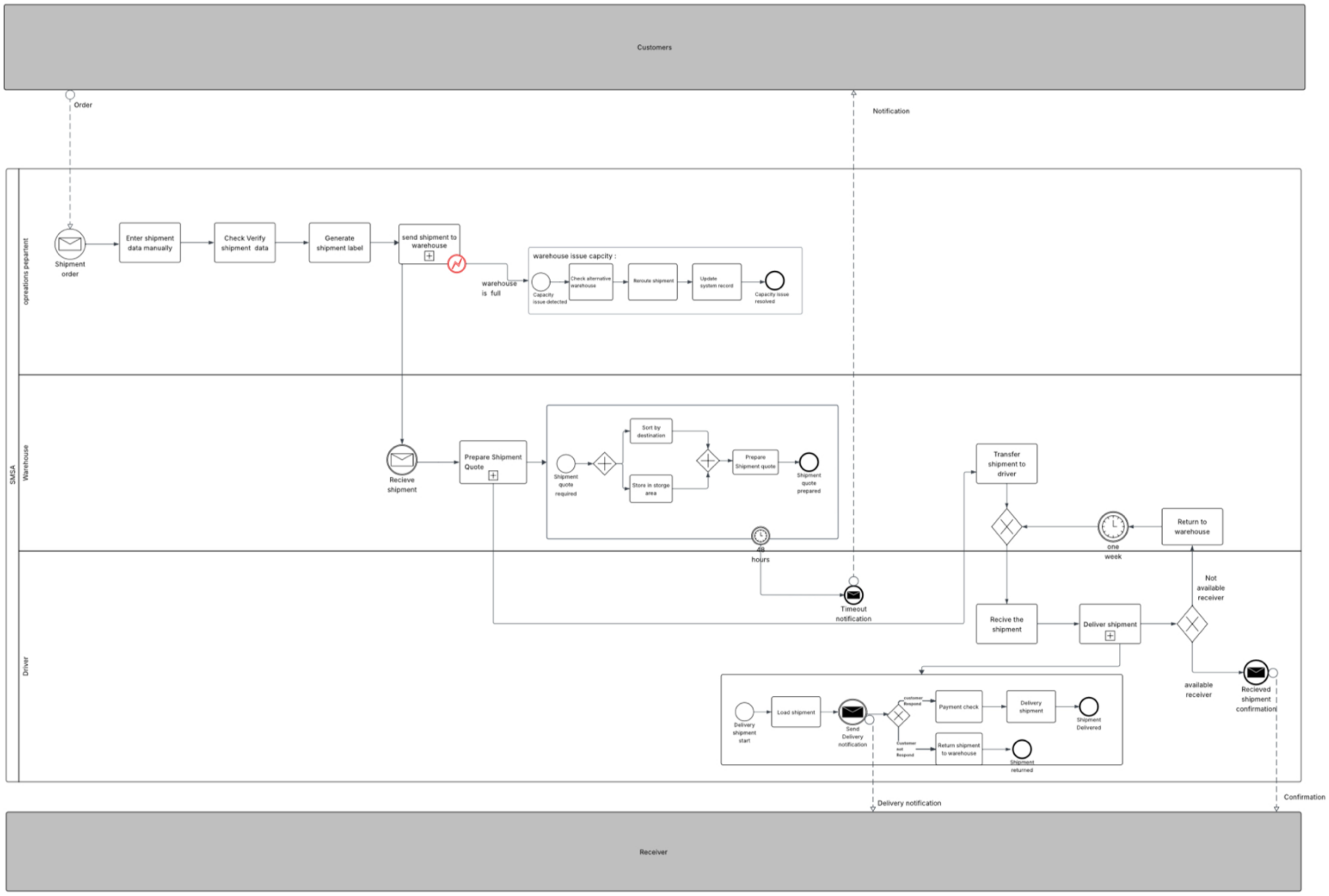
• Communication gaps and unclear instructions increase delays at the start of the delivery process.

• Manual verification steps slow down the workflow and create inconsistencies in shipment records.

• Observation confirmed that automation could significantly reduce delays and errors during dispatch.

• Peak seasons create operational bottlenecks that are not fully addressed in the current workflow.

Main process model:

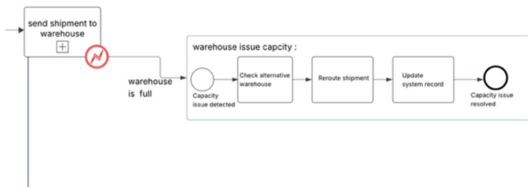


Scenario -Shipment Processing and Delivery Cycle

The process begins when the customer arrives at the SMSA branch and hands over the shipment such as a box or parcel to be sent to the recipient. Immediately after receiving the shipment, the operations department manually enters the shipment details into the system. If any required information is missing, the employee contacts the customer to request the completion of the necessary data. Once all information is provided, the employee prints the shipment label, attaches the barcode, and sends the shipment to the warehouse for further processing.When the shipment arrives at the warehouse, the warehouse team receives it and begins handling it. This includes storing the shipment inside the facility, sorting it according to the district or area, and preparing it for delivery. Once these tasks are completed, the shipment is handed over to the driver.After receiving the shipment, the driver begins the delivery journey and sends a notification to both the customer and the recipient, informing them that the shipment is on the way. When the driver reaches the destination, he checks whether the recipient is available. If the recipient is present, the shipment is delivered successfully, and the process ends. If the recipient is not available, the driver returns the shipment to the warehouse, where it is kept for one full week. After the week passes, the driver picks up the shipment again and attempts a second delivery.

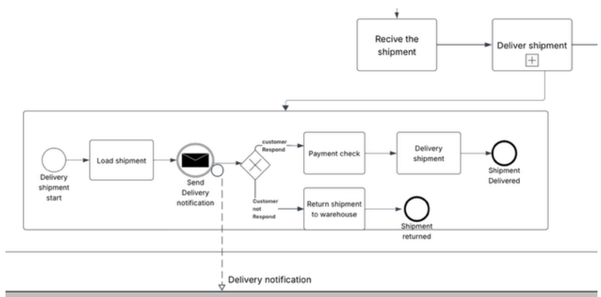
This scenario illustrates the full operational cycle of the shipment from drop-off at the branch, through warehouse processing, to final delivery and redelivery attempts.

Handling expectations and sub-processes:



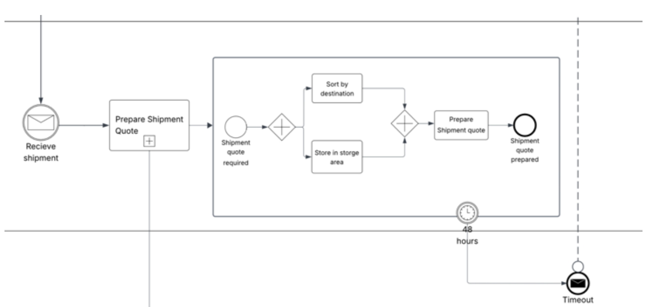
Scenario Explanation -warehouse Capacity Exception handling:

when the shipment label is generated and the shipment is sent to the warehouse, the system checks the warehouse capacity. If the warehouse is full, an exception is triggered. The employee then looks for another warehouse with available space. If an alternative warehouse is found, the shipment is rerouted, and the system updates the record. After the rerouting is completed, the process returns to the normal flow.



Scenario -Delivery Shipment Sub-Processes:

The sub-process begins with loading the shipment into the delivery vehicle. Once loaded, the system sends a delivery notification to the customer to inform them that the shipment is on its way. After the notification is delivered, the process enters a decision point where the system or driver checks whether the customer has responded. Based on the customer's response, the sub-process follows one of two possible paths: If the customer responds: The required payment check is performed (when applicable). If payment is confirmed, the shipment is delivered successfully, and the process ends with a "Shipment Delivered" end event. If the customer does not respond: The shipment is returned to the warehouse to ensure proper handling and avoid operational delays. The process then ends with a "Shipment Returned" end event.



Scenario -Prepare Shipment Quote Sub-Process:

This sub-process is added to highlight the process inside the system for preparing a shipment quote because this task involves many processes that cannot all be accommodated within one task. When this system recognize that preparing a shipment quote is needed, its contents go through one of two processes inside the system, they are sorted or held for temporary storage if they cannot be processed immediately. Both processes thus converge at the major action of preparing the shipping quote. Efficiency is ensured by placing a 48-hour timer within the sub-process. This ensures that if the quote is not completed within this period, the system automatically generates a timeout notification to avoid delay and remind personnel to act on it. The sub-process is completed once a quote is generated successfully so that the main process can move further based on actual cost and shipment details.

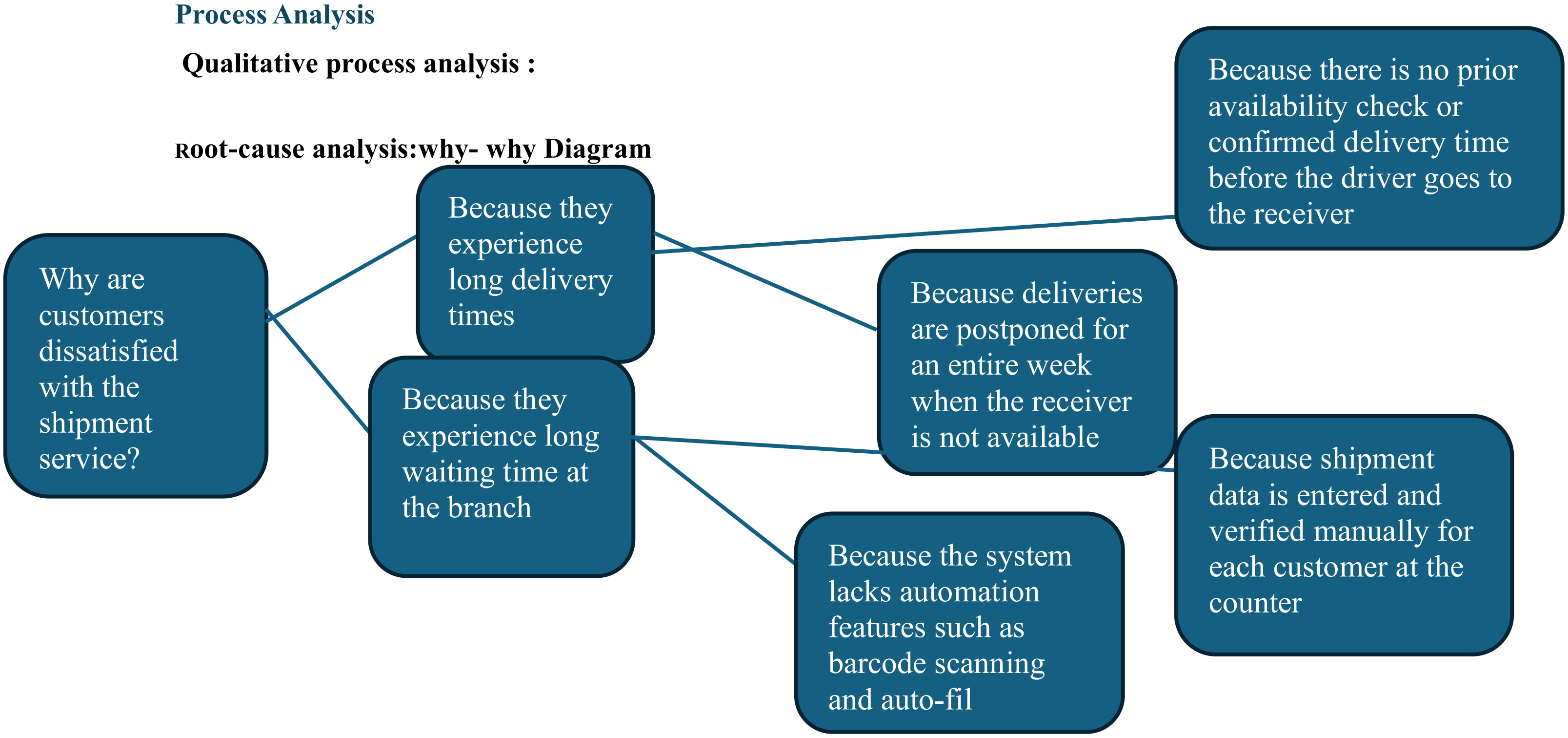
Executive Summary

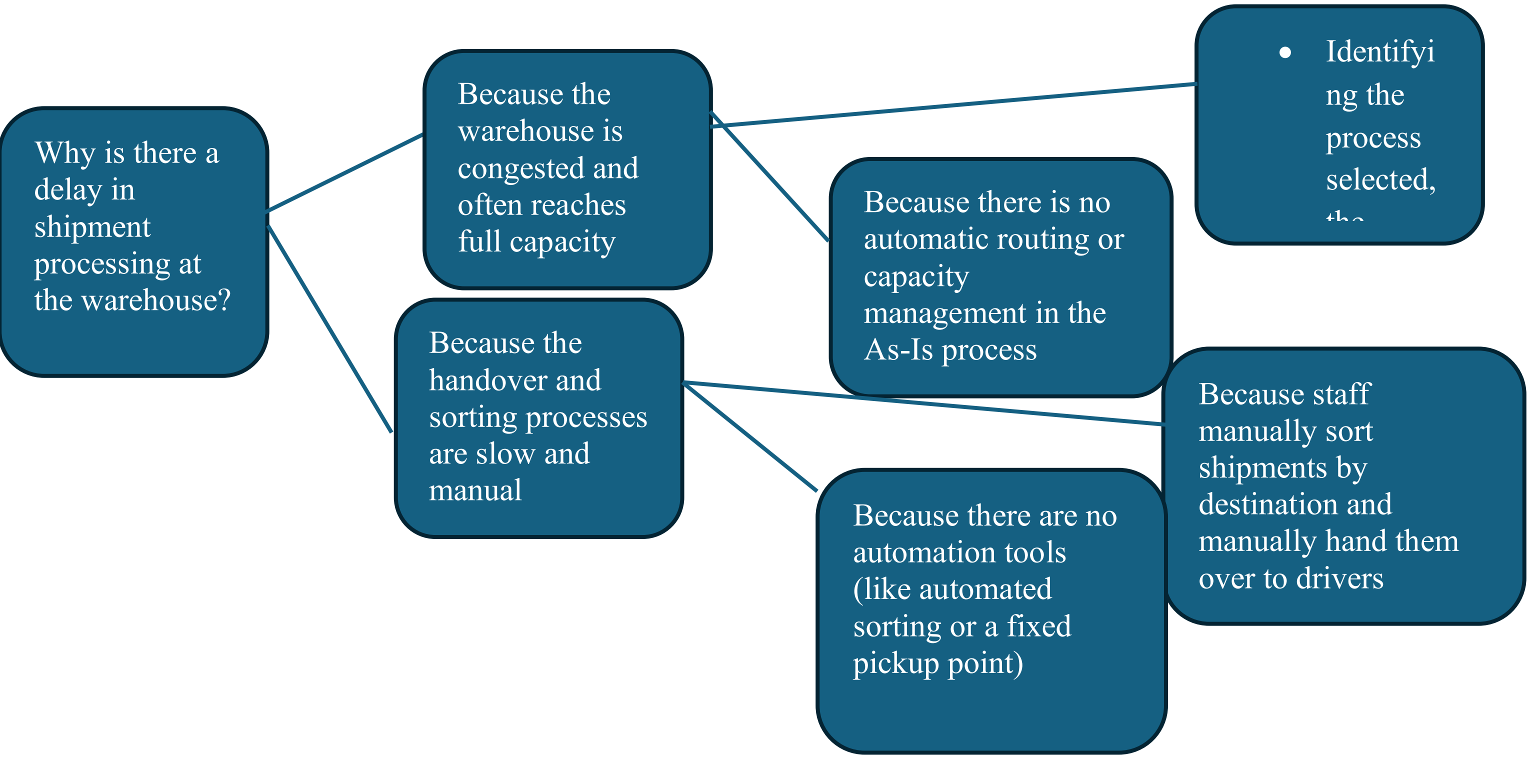
Presently, the shipment process has various issues that may hamper the speed and performance of operations and detrimentally impact customer satisfaction. With the As-Is model, there is much reliance on manual data entry, which makes errors, long queues, warehouse congestion, delayed driver handovers, and failed deliveries due to no receiver availability check probable. These problems cause queues and delays in processing and delivery. The proposed To-Be model solves these inefficiencies through the addition of automation, parallel processing during peak hours, alternative warehouse routing when capacity is full, and a receiver availability check before delivery. It also serves to reduce waiting times by improving accuracy, smoothing warehouse and driver operations, and increasing overall reliability in the delivery process. The newly redesigned process enhances the cycle time, minimizes bottlenecks, increases performance, and enhances the satisfaction of customers while strengthening the company's position in the logistic sector.

Process Analysis

Qualitative process analysis :

Root-cause analysis:why- why Diagram





(Stakeholder Analysis)

1) Customer

Role: Hands over the shipment and initiates the data-entry process.

Concerns:

- Fast data entry

- Reduced waiting time

Issue: Experiences long waiting times due to slow manual data entry.

2) Receiver

Role: Receives the shipment from the driver.

Concerns:

- Knowing the expected delivery time

- Avoiding one-week delivery delays

Issue: May not be present at delivery time because no prior notification is sent

3) Operations Department

Role: Entering shipment data into the system and printing the shipment label.

Concerns:

- High efficiency in data entry

- Completing the process quickly without delaying customers

- Reducing customer queues inside the branch

Issue:Manual data entry causes slow processing and increases customer wait times.

4) Warehouse / Sorting Center

Role: Receiving, sorting, storing, and preparing shipments for delivery.

Concerns:

- Reducing congestion

- Fast preparation of shipments

- Avoiding delays for driver

Issues: Only one warehouse serves the entire city, causing congestion and pressure and the Shipments are handed to drivers manually, causing significant delays.

5) Driver Role: Picks up shipments from the warehouse and delivers them to receivers.

Concerns:

- Quick pickup

- Avoiding long waiting times in the warehouse

Issues:Waits a long time due to manual handover and If the receiver is not available, delivery is postponed for one full week.

6) Process Owner (Operations Manager)

Role: Oversees the entire process: data entry, warehouse operations, and driver activities.

Concerns:

- Reducing process-related errors

- Ensuring compliance with procedures

- Monitoring overall performance quality

Issue: Identifies several slow steps that negatively affect service quality.

7) Process Sponsor / Executive Management

Role:

Monitors overall service quality and customer satisfaction.

Concerns:

- Increasing customer satisfaction

- Reducing complaints

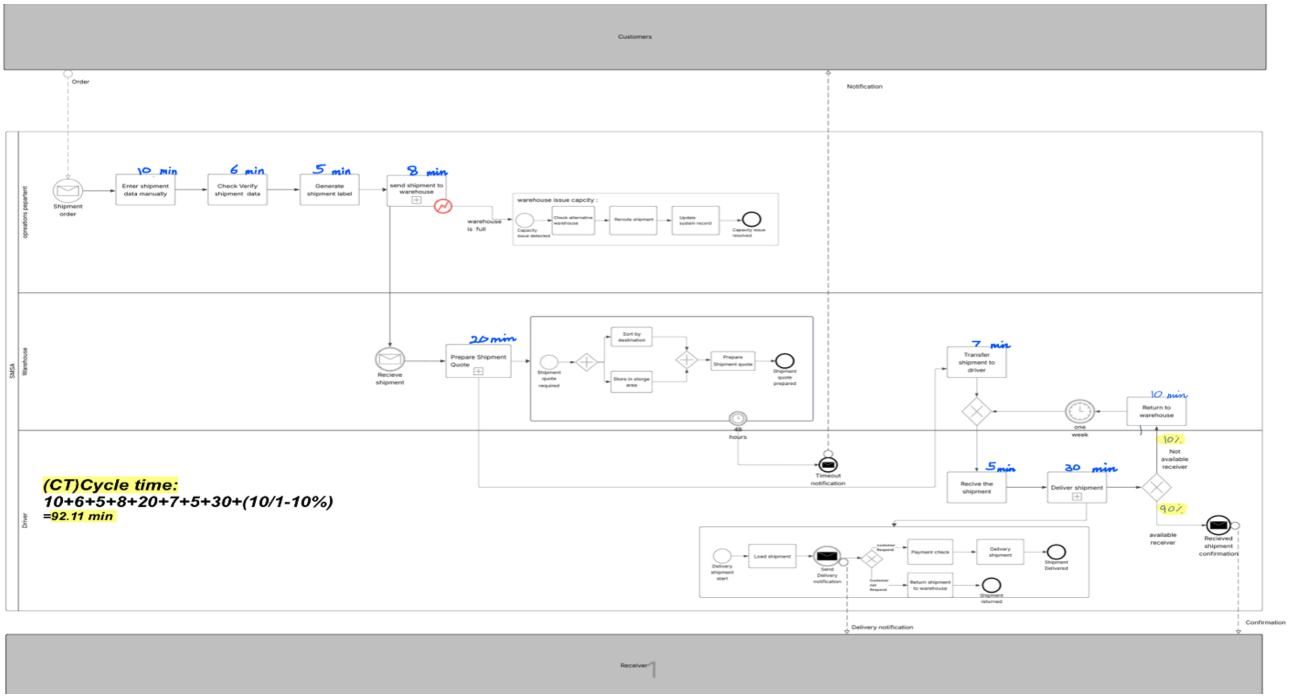
- Maintaining a strong company reputation

Issue: Delivery delays negatively affect the company’s image and increase complaints

Issue Register

Issue	Root Cause	Impact on the Process (Qualitative)	Responsible Party	Issue Classification
Slow data entry due to manual processing	No Auto-fill or Barcode tools	Long customer waiting time + delayed shipment creation	Operations Department	Operational Issue
Shipment congestion in the warehouse	Only one warehouse serves the entire city	High congestion + slow sorting + pressure on employees	Warehouse	Capacity Issue
Driver delays due to manual handover	Warehouse staff hand shipments to drivers manually instead of using a fixed pickup point	Long waiting time + late delivery start	Warehouse + Drivers	Workflow / Handover Issue
Delivery delays when the receiver is unavailable	No prior notification or confirmed delivery timing	Shipment is returned + one-week delay before reattempt	Driver + Warehouse	Delivery / Communication Issue

Quantitative processes analysis:



(Flow analysis)

Cycle time (CT)=10+6+5+8+20+7+5+30+(10/1-10%) =92.11 min

Theoretical cycle time(TCT)= 5.5+2+1.5+2.5+6+2+1.5+9+5=35 min

Cycle time efficiency(CTC)= 35/92.11=0.38 or 38% is close to that is mean 62% stands for delays, waiting, or non-value-adding activities and need to improvement by reducing waiting time.

	Activity Name	*Cycle Time (min)*
1	Enter shipment data manually	10
2	Check / verify shipment data	6
3	Generate shipment label	5
4	Send shipment to warehouse	8
5	Prepare shipment quote	20
6	Transfer shipment to driver	7
7	Receive the shipment	5
8	Deliver shipment	30
9	Return to warehouse	10

	Activity Name	*Processing Time (min)*
1	Enter shipment data manually	5.5
2	Check / verify shipment data	2
3	Generate shipment label	1.5
4	Send shipment to warehouse	2.5
5	Prepare shipment quote	6
6	Transfer shipment to driver	2
7	Receive the shipment	1.5
8	Deliver shipment	9
9	Return to warehouse	5

Process Redesign

4 Key Issues & Redesign Goals:

Issue 1: Slow Manual Data Entry

The As-Is process relies entirely on manual data entry when creating a shipment. This causes makes delays in processing and slows down the sorting and delivery stages.

Redesign Goal 1: Automate Data Entry by Using Barcode Scanning

Issue 2: • Shipments are piled in the warehouse because they are not distributed according to distance or capacity, which affects the processing speed

Redesign Goal 2: Improving the workflow after receiving the shipment by organizing the sorting steps, transferring the shipment to the truck, sending it to the branch, and then assigning it to the driver, in order to reduce delays and speed up the processing and delivery process.

Issue 3: Single-Point Dependency (Only One Employee Handling All Data Entry)

The process depends on one employee who performs all entry tasks. This creates long queues, high workload pressure, and vulnerability.

Redesign Goal 3: Introduce Parallel Processing or Task Splitting

This could include adding a second service point during peak times or splitting simple tasks to another employee.

Issue 4: Lack of Real-Time Visibility for Drivers and Operations Team

Drivers wait without knowing when a shipment will be ready. This increases waiting time and delays deliveries.

Redesign Goal 4: Provide Real-Time Status Updates to Drivers and Add a “Ready for Dispatch” status or notification to inform drivers when the shipment is fully prepared.

Heuristics

1– Time Heuristic: Automation of Shipment Preparation

Problem:

In the As-Is process diagram, the employee performs several sequential manual steps:

- Enter shipment data manually
- Check/Verify shipment data
- Generate shipment label

These steps slow down the processing time because:

- They depend on one employee
- Each step requires manual typing
- Any missing information forces re-verification
- High chance of errors during peak periods

Redesign Solution:

Automate shipment preparation using barcode scanning and auto-generated invoice/label data.
The employee no longer types details; instead:

- Scan shipment barcode
- System automatically retrieves customer information

- Weight, size, and distance measurements feed directly into invoice generation
- System generates invoice + shipment label instantly

This fully replaces the multi-step manual preparation in the As-Is model.

How This Will Improve the Process:

- Eliminates the slowest part of the As-Is workflow
- Reduces time between Order → Shipment sent to warehouse
- Reduces manual errors
- Ensures shipments move to sorting faster
- Decreases customer waiting time at the counter
- Allows warehouse operations to begin earlier, reducing end-to-end cycle time

2– Capacity Heuristic: Parallelism in Data Verification & Invoice Preparation

Problem:

- In the As-Is model, all preparation tasks are handled by one employee only:
- Data entry
- Data verification
- Label generation
- Sending shipment to warehouse

This creates a single-point bottleneck, especially when:

- More than one customer arrives
- Manual data verification takes too long
- Shipments pile up waiting for labels
- Warehouse receives shipments late
- There is no support role in the As-Is process.

Redesign Solution:

Main employee does:

1. Invoice generation
2. Label printing
3. System updates

Supporting employee simultaneously performs:

1. Weight measurement
2. Size measurement
3. Basic identity/receiver info checking

How This Will Improve the Process:

- Removes the bottleneck that existed in the As-Is model
- Speeds up invoice and label preparation

- Reduces customer queue length
- Ensures shipments reach the warehouse without delay
- Increases processing capacity during high demand
- Maintains service quality by distributing workload

3– Information Transparency Heuristic: Real-Time Shipment Readiness Notifications

Problem:

- Drivers often rely on manual updates
- They wait without knowing whether shipments are “Prepared”, “Sorted”, or “Ready for delivery”
- Communication between warehouse and drivers is not synchronized
- Drivers leave late due to waiting for confirmation

This results in:

1. Idle time
2. Slow dispatch
3. Delayed deliveries
4. Inefficient coordination between lanes

Redesign Solution:

Once sorting is completed (Sort by destination), the system sends
“Receiver availability check” and delivery notifications to driver and customer.

Shipment readiness becomes visible immediately through system events (dotted message flows).

Therefore, the heuristic redesign is:

Add automated “Shipment Ready for Dispatch” notifications sent from the system to the assigned driver and operations team as soon as sorting is complete.

These notifications include:

- Shipment readiness
- Sorting completion
- Driver assignment
- Receiver availability response

Link Between Issues, Redesign Goals, and Applied Heuristics

The key issues, redesign goals, and redesign heuristics (presented in separate sections) are directly connected. Issues identified through the As-Is analysis guided the choice of appropriate redesign goals. Subsequently, each goal was linked to one of the heuristics applied.

Problem: Manual data entry causes delays →Redesign Goal:

Reduce preparation time → Heuristic Applied: Automation of tasks.

Automating data entry through barcode scanning and auto-fill directly removes the slowest step identified in the process.

Problem: Queue formation and workload bottlenecks → Redesign Goal: Reduce waiting time and balance workflow → Heuristic Applied: Parallelism (Perform tasks simultaneously).

Breaking simple verification steps into parallel to run alongside the main check decreases waiting lines and avoids congestion.

Problem: Lack of shipment status visibility → Redesign Goal: Enhance communication and transparency →Heuristic Applied: Information Transparency.

Real-time system notifications ensure departments and customers receive instantaneous updates, thereby reducing delays on account of manual follow-ups.

These links show how each heuristic directly addresses the underlying problems in the As-Is model and supports the redesign goals reflected in the To-Be process.

Alternative Redesign Options Considered but Not Adopted

A number of redesign options were considered but ultimately not adopted in the To-Be model

- outsourcing the preparation or sorting activities of the first shipment to third-party logistics providers , this option was discarded since that would diminish SMSA's control over operations and create dependency risks.

-Redesign the As-Is process by combining invoice generation ,scanning of barcodes ,and measuring in one single activity .It was not adopted because if too many steps are combined into one ,human error increases and reduces clarity in process.

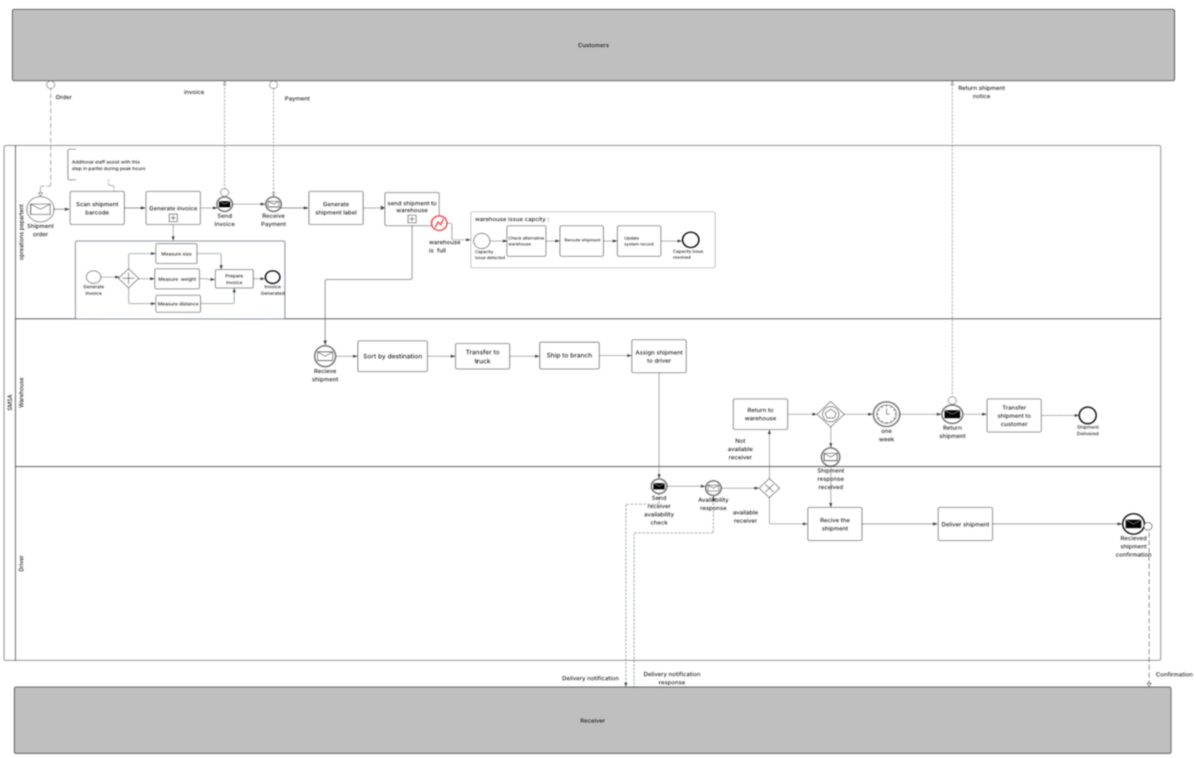
-The implementation of a full end-to-end automated conveyor for sorting was considered but has been dismissed due to high infrastructure costs and major physical changes that would be required in the warehouse..

Therefore ,the chosen redesign concentrated on viable system-based enhancements such as automation of data entry ,parallel support in peak periods ,and increased transparency in tracking without pursuing operationally disruptive or economically unviable change

Overview of the Redesigned-To-Be Process Model

The redesigned To-Be model introduces several targeted improvements which directly address issues identified in the As-Is process .Manual data entry and shipment preparation are replaced by barcode scanning and system-generated information ,which eradicates the slowest and most error-prone activity of the original workflow .More importantly ,parallel execution is introduced into this model .During peak hours ,an assisting staff member will be called to participate in the process for support in verification and invoice preparation .This can reduce reliance on a single employee and queue formation .System-driven notifications for shipment readiness ,receiver availability checks ,and delivery updates are also integrated into the To-Be process .Such real-time transparency will enable faster coordination among the warehouse ,drivers ,and customers .These modifications combined create a much quicker ,more reliable , better synchronised workflow ,that meets the redesign goals and the root causes of inefficiency identified from the As-Is model

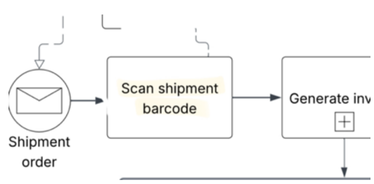
“To-Be” Model



Key Modifications and Improvements Introduced in the To-Be Model

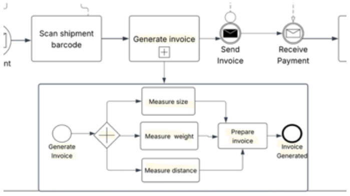
1. Replacement of Manual Data Entry with Barcode Scanning

The manual shipment data entry step was replaced with an automated “Scan shipment barcode” activity to eliminate typing delays and reduce human error.



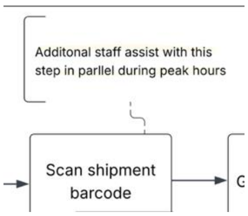
2. Parallel Execution of Measurement & Verification Tasks (Instead of Sequential Work)

The To-Be model restructures measurement activities (weight, size, distance) and verification steps into parallel tasks, instead of performing everything sequentially under one employee.
This eliminates the bottleneck caused by single-point dependency in the As-Is model.



3. Additional Staff Support During Peak Hours (Parallelism Heuristic)

A parallel assisting-staff was illustrated so a second employee can participate during peak hours. This reduces workload concentration and prevents queue buildup.

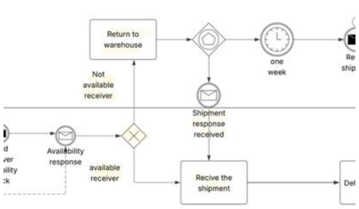


4. Introduction of Real-Time System Notifications (Information Transparency Heuristic)

The To-Be model adds automatic system messages for:
- shipment readiness
- receiver availability
- delivery updates

These updates reduce delays caused by waiting for manual follow-ups.

5. Addition of New Decision Gateways for Automated Flow Control

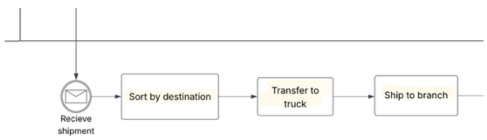


New gateways were introduced such as “Receiver available?” and “Shipment response received” to route shipments automatically based on availability and customer response.

6. Enhanced Sorting for Better Warehouse Distribution

Shipments are now sorted and routed to the correct warehouse or branch based on destination and capacity conditions, reducing congestion and preventing delays caused by warehouse overload.

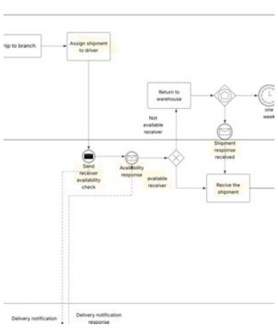
7. Removal of Redundant Manual Checks Present in the As-Is Model



Repeated manual verification steps were removed or consolidated, reducing unnecessary rework and shortening preparation time.

8. Improved Driver Assignment Through Automated Availability Confirmation

The To-Be model introduces system-generated receiver availability checks immediately after driver assignment. These automatic notifications ensure drivers are informed in real time about whether the receiver is available, minimizing unnecessary travel, reducing idle time, and enabling deliveries to proceed only when conditions are confirmed. This redesign improves coordination, reduces delays, and accelerates final delivery



Key findings:

- **1-**The process became clearer and easier to control and monitor
- **2-**Customer satisfaction levels increased
- 3-Congestion and bottlenecks were reduced
- 4-Communication and coordination between departments improved

Conclusion:

In conclusionThe redesigned process model aims to address the operational challenges and difficulties that previously affected performance efficiency and customer experience at SMSA. Through analyzing the (As-Is) process and identifying weaknesses such as slow manual data entry and warehouse congestion the team was able to develop a more effective and flexible (To-Be) model.

The quantitative analysis, including cycle time calculation and process efficiency assessment, helped in designing improvements that reduce waste and accelerate shipment processing. Additionally, incorporating Rainy Day scenarios enhanced SMSA's readiness to handle real-world issues such as warehouse overloads and shipment delays

Overall, the improved process model enhances SMSA's operational performance by speeding up shipment handling, clarifying task distribution, and optimizing resource utilization. These improvements support better service quality and higher customer satisfaction It also contributes to strengthening SMSA's ability to compete and adapt within the rapidly evolving logistics and delivery sector in Saudi Arabia.

References :

- Home | SMSA Express. (n.d.). https://www.smsaexpress.com/sa/
- Dumas, M., La Rosa, M., Mendling, J., & Reijers, H. A. (2018). Fundamentals of Business Process Management. Springer.